

a continuation-based solution of the linearity challenge

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what is the linearity challenge?

Carbone et al., 2024

The Concurrent Calculi Formalisation Benchmark

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- collection of challenges on **mechanisation** of process calculi
- linearity challenge = linear management of session endpoints

why is it a challenge?

$$\frac{P \vdash \Gamma \quad Q \vdash \Delta}{P \mid Q \vdash \Gamma, \Delta}$$

State of affairs

- context splitting is a **source of complexity**
- look for techniques to **avoid/simplify context splitting**
 - **leftovers** Allais, 2017; Zalakain and Dardha, 2021
 - **linearity predicates** Sano, Kavanagh, and Pientka, 2023
 - **tagged contexts** Zackon et al., 2025

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Our twist to the challenge

- **embrace context splitting**
- find a model for which context splitting “works well”

LCC: a session model based on linear logic...

Caires and Pfenning, 2010; Wadler, 2014

$$\frac{P \vdash \Gamma, x : A \quad Q \vdash \Delta, x : A^\perp}{(x)(P \mid Q) \vdash \Gamma, \Delta}$$

- a thread cannot own both endpoints of a session **by design**
- no need for **polarities** x^+, x^- Gay and Hole, 2005
- no need for **double binders** $(\nu xy) \dots$ Vasconcelos, 2012

...with explicit continuations

Abramsky, 1993; Bellin and Scott, 1994; Dardha, Giachino, and Sangiorgi, 2017

CP

updated

$$\frac{P \vdash \Gamma, y : A \quad Q \vdash \Delta, x : B}{x[y](P \mid Q) \vdash \Gamma, \Delta, x : A \otimes B}$$

...with explicit continuations

Abramsky, 1993; Bellin and Scott, 1994; Dardha, Giachino, and Sangiorgi, 2017

CP

updated

$$\frac{P \vdash \Gamma, y : A \quad Q \vdash \Delta, \textcolor{red}{x} : B}{x[y](P \mid Q) \vdash \Gamma, \Delta, \textcolor{red}{x} : A \otimes B}$$

LCC

added

$$\frac{P \vdash \Gamma, y : A \quad Q \vdash \Delta, \textcolor{red}{z} : B}{x[y, \textcolor{red}{z}](P \mid Q) \vdash \Gamma, \Delta, \textcolor{red}{x} : A \otimes B}$$

fresh continuation

removed

- session = chain of linear channels
- no need for **updating** the type of channels
- “removing + adding” **easier** than “updating”

representation of typing contexts

Context : Set

Context = List Type

Use co-de Bruijn indices (McBride, 2018)

- channels are **nameless**
- the **order of types** is consistent with the order of binders
 - first type = newest channel
 - “removing + adding” easier than “updating”
- only **relevant channels** have their type in the context

context splitting = order preserving partitioning

data $_ \simeq _ + _ : \text{Context} \rightarrow \text{Context} \rightarrow \text{Context} \rightarrow \text{Set}$ where

• $: [] \simeq [] + []$

$< _ : \forall \{A \Gamma \Delta \Theta\} \rightarrow \Gamma \simeq \Delta + \Theta \rightarrow A :: \Gamma \simeq A :: \Delta + \Theta$

$> _ : \forall \{A \Gamma \Delta \Theta\} \rightarrow \Gamma \simeq \Delta + \Theta \rightarrow A :: \Gamma \simeq \Delta + A :: \Theta$

Example: separate positive from negative constants

pos-neg : $(\perp :: \top :: 0 :: 1 :: []) \simeq (0 :: 1 :: []) + (\perp :: \top :: [])$

pos-neg = $> > < < \bullet$

Example: separate multiplicative from additive constants

mul-add : $(\perp :: \top :: 0 :: 1 :: []) \simeq (\perp :: 1 :: []) + (\top :: 0 :: [])$

mul-add = $< > > < \bullet$

channels/processes as predicates on contexts

Rouvoet et al., 2020

```
data Ch (A : Type) : Context → Set where  
  ch : Ch A [ A ]
```

channels/processes as predicates on contexts

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data Ch (A : Type) : Context → Set where  
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```
data Proc : Context → Set where
```

$$\frac{}{\text{close } x \vdash x : \mathbf{1}}$$

```
close :  $\forall [ \text{Ch } \mathbf{1} \Rightarrow \text{Proc} ]$ 
```

- $\forall [\dots]$ is **universal quantification** over contexts

some help from separation logic

$$\frac{P \vdash \Gamma}{\text{wait } x.P \vdash \Gamma, x : \perp}$$

$$\text{wait} : \forall [\text{Ch } \perp * \text{Proc} \Rightarrow \text{Proc}]$$

- $*$ is **separating conjunction** (O'Hearn and Pym, 1999)
- $P * Q$ holds for Γ if
 - $\Gamma \simeq \Delta + \Theta$
 - P holds for Δ
 - Q holds for Θ

separating conjunctions galore!

$$\frac{P \vdash \Gamma, y : A \quad Q \vdash \Delta, z : B}{x[y, z](P \mid Q) \vdash \Gamma, \Delta, x : A \otimes B}$$

$\text{fork} : \forall \{A \ B\} \rightarrow \forall [\text{Ch } (A \otimes B) * \\ ((A :: _) \vdash \text{Proc}) * \\ ((B :: _) \vdash \text{Proc}) \Rightarrow \text{Proc}]$

- all context splittings **hidden** by $*$ in the syntax...
- ...but they **resurface** in the semantics



a jungle of mechanisations

warning: comparing oranges, apples, bananas...

	Model	Linearity	Channels	Sub. red. (Kb)
Goto et al. (2016)	$S\pi$	splits	loc. nameless	543
Castro-Perez, Ferreira, and Yoshida (2020)	$S\pi$	splits	loc. nameless	204
Thiemann (2019)	GV	splits	co-de Bruijn	177
Zackon et al. (2025)	CP	tags	HOAS	129
Zalakain and Dardha (2021)	$L\pi$	leftovers	de Bruijn	90
Rouvoet et al. (2020)	GV	splits	co-de Bruijn	82
Ciccone and Padovani (2020)	$L\pi$	splits	co-de Bruijn	77
Jacobs, Balzer, and Krebbers (2022)	GV	splits	named	68
Sano, Kavanagh, and Pientka (2023)	SCP	predicates	HOAS	35
this talk	LCC	splits	co-de Bruijn	21

- context splitting is found across the whole range of sizes

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- context splitting is found across the whole range of sizes
- GV primitives consume and return (linear) channels: **continuations!**
- Sano, Kavanagh, and Pientka, 2023 rename channels after every usage: **continuations!**

summary

Achievements

- small mechanisation of (continuation-based) session model
- readable (typed) syntax (kudos to Rouvoet et al., 2020)
- extensible base (see repo)
 - ✓ deadlock freedom (Jacobs, Balzer, and Krebbers, 2022 is **10x**)
 - ✓ context-free session types

Drawbacks

- not so readable (typed) semantics

Observation

- is context splitting that bad? **Not necessarily!**

thank you



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